

Query Processing – DSA 0501

1. Case Study: Monthly Hospital Patient Visits

A hospital administrator wants to analyze the number of patients visiting the hospital each month during 2025.

Month

Jan

Patient

Visits

Jan 1200

Feb 1350

Mar 1500

Apr 1450

May 1700

Jun 1850

Questions

1. Which visualization is most suitable?

Answer:- Line Chart (Shows the trend over time).

2. What trend can be observed?

Answer:- The number of patients generally increases from January to June, with slight dip in April.

3. Identify the month with the highest patient count.

Answer:- June (1850 patients)

4. Predict whether visits are increasing or decreasing.

Answer:- The visits are increasing overall.

2. Case Study: Study Hours vs Exam Marks

A teacher wants to determine whether studying more hours improves exam performance.

Student

Study

Hours

Mark

Score

1

A 2 45

B 3 52

C 4 58

D 5 68

E 7 82

Questions

1. Which chart is appropriate?

Answer:- Scatter Plot

2. Is there a positive or negative relationship?

Answer:-Positive relationship (more study hours lead to higher marks)

3. Which student is an outlier, if any?

Answers:- No significant outlier is present.

4. Can marks be predicted from study hours?

Answer:- Yes, because there is a positive correlation.

3. Case Study: Employee Salary Distribution

A company wants to understand how salaries are distributed among employees.

Salary Data (₹ in Thousands)

25, 28, 30, 32, 35, 35, 36, 38, 40, 42, 45, 48, 50, 52, 55, 60, 65, 70

Questions

1. Which graph should be used?

Answer:- Histogram

2. In which salary range do most employees fall?

Answer:- Approximately ₹30,000 - ₹50,000

3. Is the distribution symmetric or skewed?

Answer:- Slightly right-skewed (positively skewed).

4. Are there any salary ranges with very few employees?

Answer:- ₹60,000 - ₹70,000 has comparatively fewer employees.

4. Case Study: Favorite Programming Language

A survey was conducted among 200 students.

Language

e

Number of

Students

Python 80

Java 50

C++ 30

JavaScript

t 25

R 15

Questions

1. Which visualization is suitable?

Answer:- Bar Chart

2. Which language is most popular?

Answer:- Python (80 students)

3. Which language is least preferred?

Answer:- R (15 students)

4. Compare Python and Java popularity.

Answer:-

- Python has 80 students, Java has 50 students
- Python is preferred by 30 more students than Java.

5. Case Study: Household Monthly Expenses

A family spends money in the following categories:

Category	Expense
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(%)	
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Rent	40
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Food	25
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Education	15
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Transportation	
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10	
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Entertainment 10

Questions

1. Which chart best represents the data?

Answer:- Pie Chart

2. Which category consumes the largest budget?

Answer:- Rent (40%)

3. What percentage is spent on Food and Education together?

Answer:- $25\% + 15\% = 40\%$

4. Which categories occupy the smallest share?

Answer:- Transportation (10%) and Entertainment(10%)

6. Case Study: Student Test Scores

Scores obtained by students:

45, 50, 52, 55, 58, 60, 62, 65, 68, 70, 72, 75, 78, 80, 95

Questions

1. Which plot should be used?

- **Box Plot**

2. What is the median score?

- Since there are **15 scores**, the median is the **8th value**.
- **Median = 65**

3. Are there any outliers?

- **No.** Using the IQR method:
 - $Q1 = 55$
 - $Q3 = 75$
 - $IQR = 20$
 - Upper fence = $75 + (1.5 \times 20) = 105$
 - Lower fence = $55 - (1.5 \times 20) = 25$
 - All scores (45 to 95) lie within these limits.

4. What is the spread of the data?

- **Minimum score = 45**
- **Maximum score = 95**
- **Range = $95 - 45 = 50$**
- **IQR = 20**